

PRANAV JINDAL

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EDUCATION

Bachelor of Technology - Computer Science and Engineering
Vellore Institute of Technology

Graduated **June, 2025**
8.22 CGPA

TECHNICAL SKILLS

Programming: Python, C++, SQL, PySpark

AI & ML: Scikit-learn, XGBoost, Deep Learning (CNN, YOLO), Recommender Systems, Anomaly Detection

Data Science: Statistics, Hypothesis Testing, Data Preprocessing, Feature Engineering

Model Evaluation: ROC-AUC, Precision, Recall, F1, Confusion Matrix

Cloud & DevOps: AWS (Certified), GCP, Docker, CI/CD pipelines

Tools: Git, VS Code, IntelliJ IDEA

WORK EXPERIENCE

Indian Institute of Technology, Bombay: Project Research Assistant [On-Site] **October 2025 – Present**

- Working on 5G/6G core network systems, including building a full 5G standalone testbed using Open5GS and OpenAirInterface and analyzing AMF internals, signalling flows, and SBI interactions. Contributing to next-generation 6G architecture research focused on scalable, service-driven signalling and control-plane redesign.

Indian Institute of Technology, Roorkee: Project Research Associate [On-Site] **August 2025 – September 2025**

- Collaborated with different teams to align traffic analytics solutions and deploy real-time VA-ANPR systems.
- Designed algorithms to improve accuracy across diverse traffic conditions and challenging lighting scenarios.
- Implemented automatic camera calibration using Depth Anything model, reducing setup time and ensuring alignment.
- Optimized ADAS performance on Jetson by fine-tuning Clearnet and YOLO models for faster, more reliable object recognition.

Centre for Artificial Intelligence and Robotics, DRDO, Bangalore: Project Trainee [On-Site] **March, 2025 – May 2025**

- Delivered cost-effective and scalable mapping tools for agriculture, urban planning, and defense applications.
- Led implementation of an image segmentation pipeline using GeoSAM for remote sensing under senior scientific guidance.
- Designed a robust Python solution to process satellite imagery on CPUs, reducing GPU and GIS dependencies.
- Extracted high-accuracy segmentation masks and geospatial data using Rasterio, Shapely, and Leafmap libraries.

SVGS IT Solutions, Noida: Software Engineer Intern [Hybrid] **May 2024 – November 2024**

- Designed and deployed ML pipelines into production, delivering actionable business insights across teams.
- Created interactive dashboards to effectively communicate key data insights and model performance to stakeholders.
- Enhanced model accuracy and efficiency by refining algorithms and workflows based on comprehensive data analysis.
- Developed and optimized ML models through preprocessing, feature engineering, and evaluation using Python libraries.

PROJECTS

Harvest Watch - IoT and ML for Agriculture April, 2025

Integrated IoT and ML for real-time disease detection, empowered farmers via Flutter app, ensured cost-effectiveness, and fostered community engagement.

- Developed and trained a CNN (Inception_V3) on the PlantVillage dataset (54K+ images) for plant disease classification.
- Optimized performance via preprocessing, augmentation, and hyperparameter tuning; converted to TFLite for mobile use.
- Integrated the TFLite model into an Android app for real-time, on-device disease detection to support farmers.
- Validated model accuracy using test-time augmentation and cross-validation; created summary reports.

Customer Churn Prediction Using Machine Learning November, 2024

Built ML models to predict churn using the Telco dataset, uncovered key churn factors via EDA, and improved minority class recall with SMOTE.

- Preprocessed data by handling nulls, encoding categorical variables, and applying SMOTE to balance class distribution.
- Explored customer behavior using visualizations and correlation analysis to identify churn-driving factors.
- Trained and evaluated models (Logistic Regression, Random Forest, XGBoost) with high precision and F1-score.
- Evaluated model using confusion matrix, precision-recall curve, ROC-AUC; interpreted outputs for risk-driven decisioning.

OTHER WORK EXPERIENCE

Co-Lead at Data Science Club VITB, : June, 2023

- Coordinated 5+ events with a 25-member team, increasing member engagement by 70%.
- Mentored 20+ juniors on ML projects and conducted Python/ML workshops, cultivating a strong data science culture.